Chapter 7: Transport Layer



Introduction to Networking

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Application	
Presentation	
Session	**Sessions**
Transport	**Segments, TCP/UDP, Windowing, ACK, Seq#, Ports Flow control**
Network	**Packets, Router/Multilayer Switch, IP**
Data Link	**Frames, Switch/Bridge, MAC**
Physical	**Bits, Hub/Repeater**

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Role of the Transport Layer

The transport layer is responsible for establishing a temporary communication session between two applications and delivering data between them.

TCP/IP uses two protocols to achieve this:

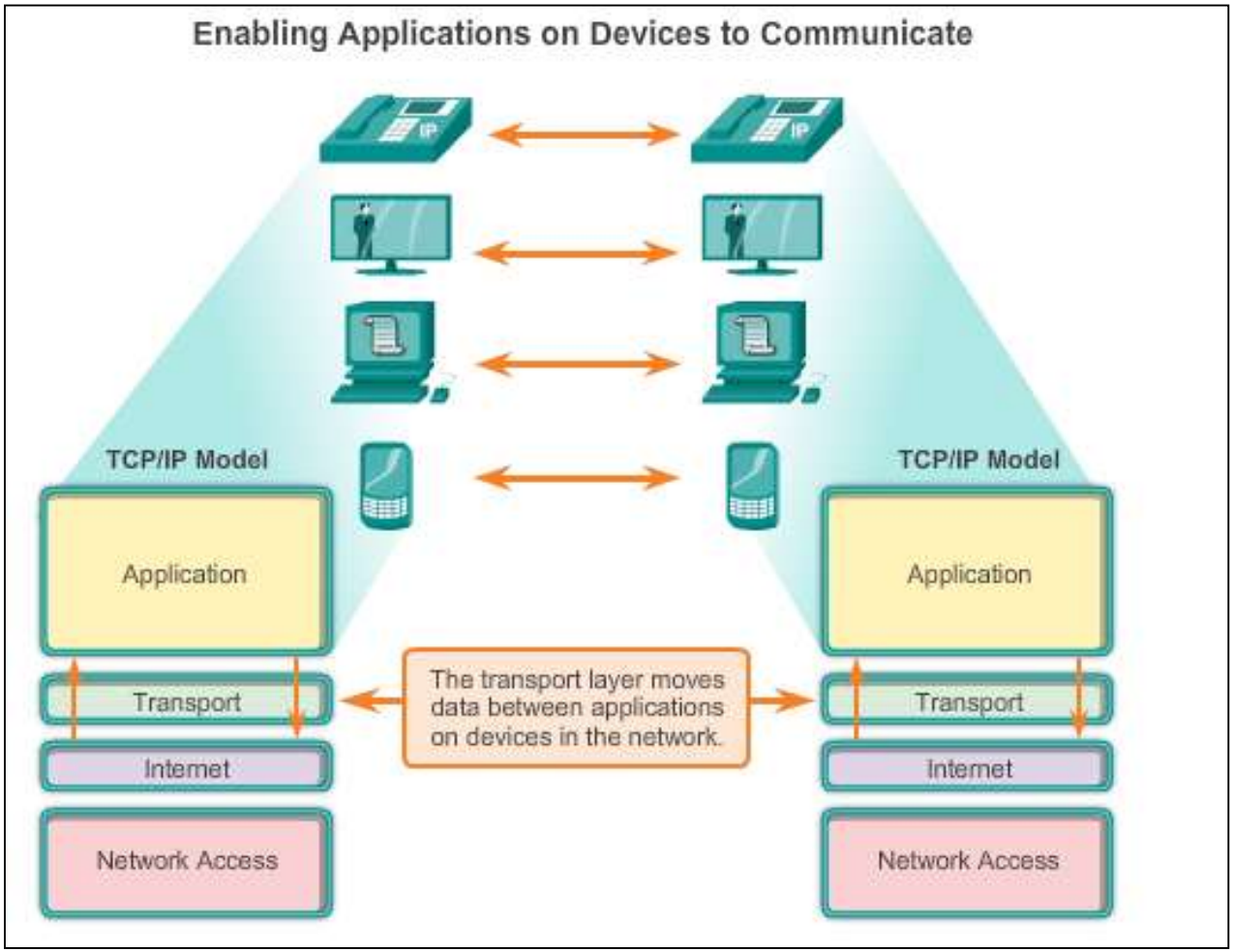
- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)

Primary Responsibilities of Transport Layer Protocols

- Tracking the individual communication between applications on the source and destination hosts
- Segmenting data for manageability and reassembling segmented data into streams of application data at the destination
- Identifying the proper application for each communication stream

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Role of the Transport Layer (Cont.)



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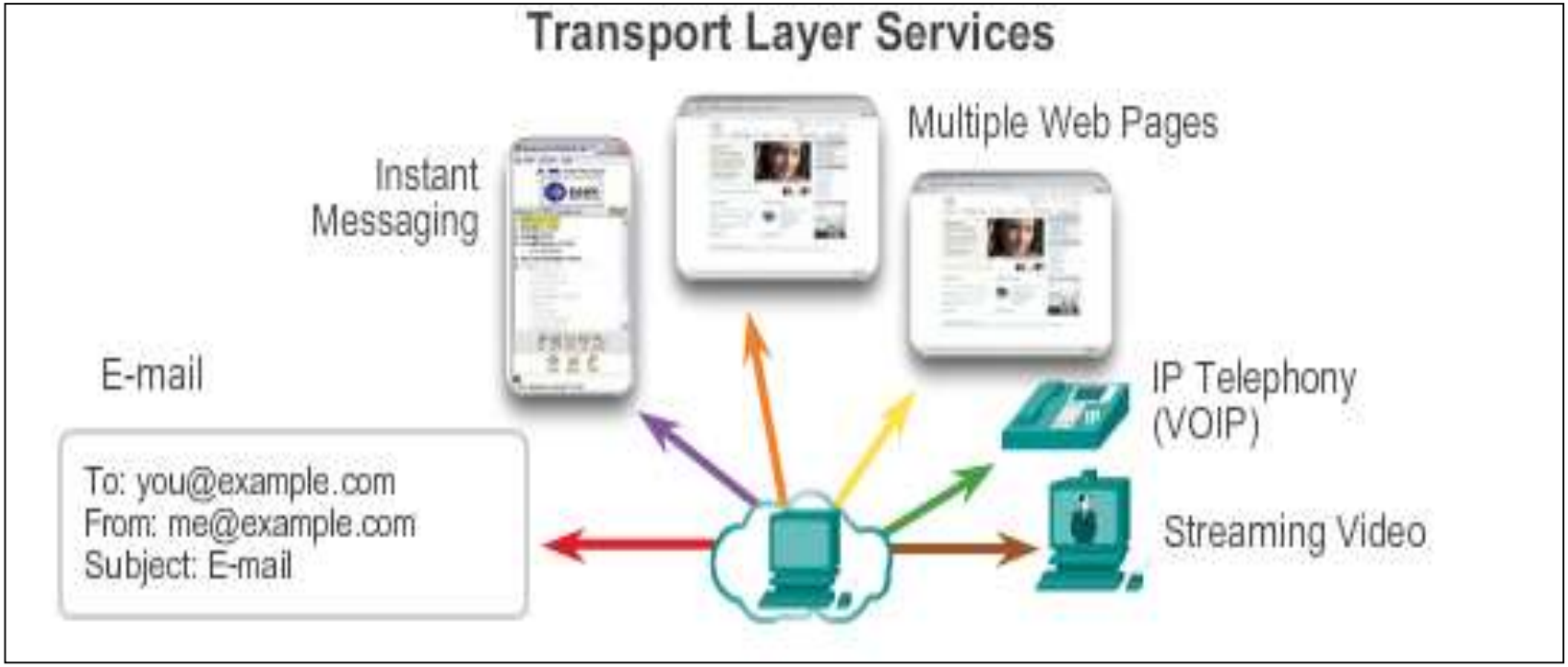
Conversation Multiplexing

Segmenting the Data

- Enables many different communications, from many different users, to be interleaved (multiplexed) on the same network, at the same time.
- Provides the means to both send and receive data when running multiple applications.
- Header added to each segment to identify it.

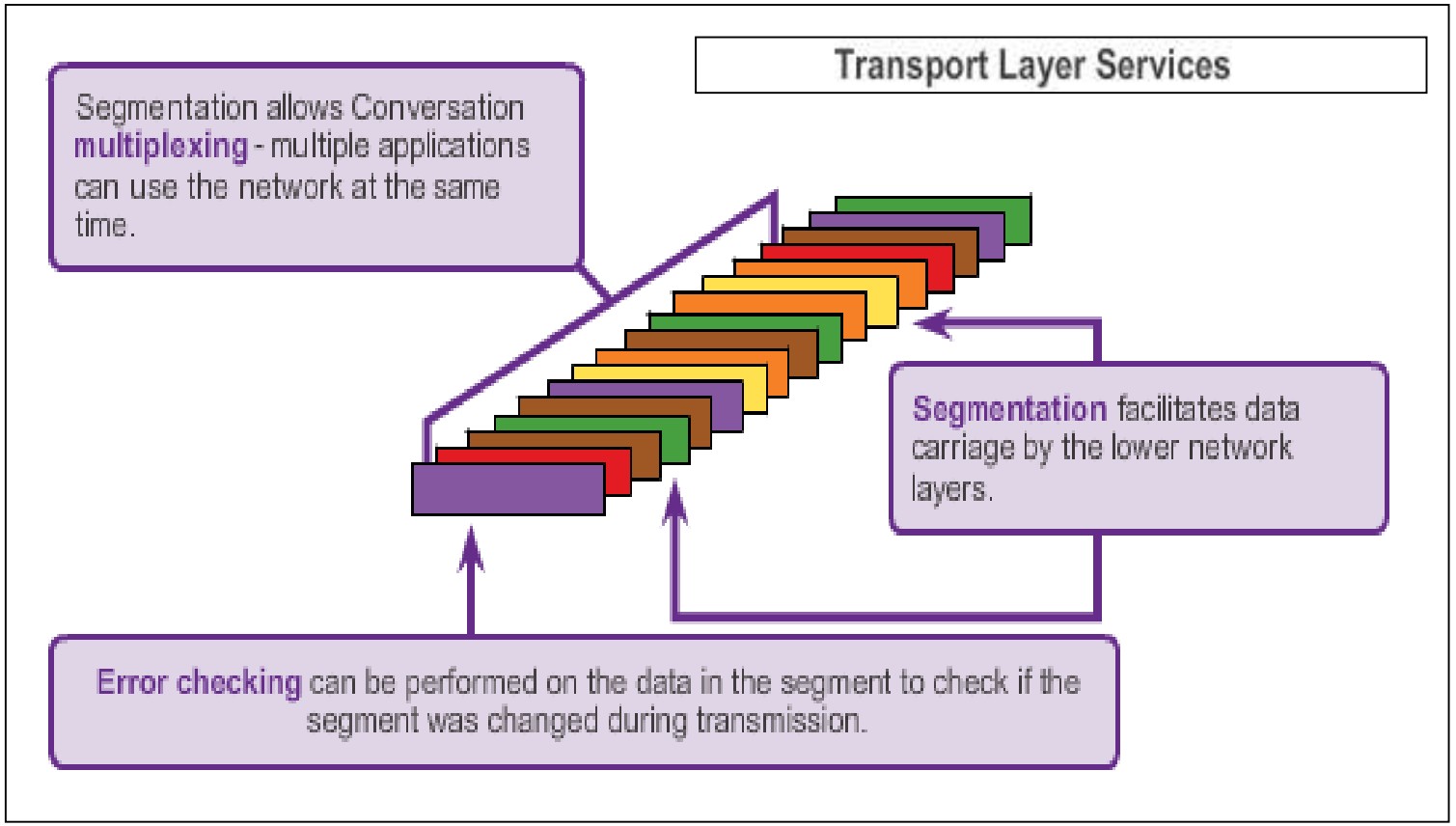
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Conversation Multiplexing (Cont.)



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Conversation Multiplexing (Cont.)



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Transport Layer Reliability

Different applications have different transport reliability requirements.

TCP/IP provides two transport layer protocols, **TCP and UDP.**

TCP

- Provides reliable delivery ensuring that all of the data arrives at the destination.
- Uses acknowledged delivery and other processes to ensure delivery
- Makes larger demands on the network – more overhead.

UDP

- Provides just the basic functions for delivery – no reliability.
- Less overhead.

TCP or UDP

- There is a trade-off between the value of reliability and the burden it places on the network.
- Application developers choose the transport protocol based on the requirements of their applications.

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Introducing TCP

- Defined in RFC 793
- Connection-oriented – Creates a session between the source and destination
- Reliable delivery – Retransmits lost or corrupt data
- Ordered data reconstruction – Reconstructs numbering and sequencing of segments
- Flow control – Regulates the amount of data transmitted
- Stateful protocol – Tracks the session

TCP Segment

Bit(0)	Bit(15)	Bit(16)	Bit(31)	
Source Port (16)		Destination Port (16)		20 Bytes
Sequence Number (32)				
Acknowledgement Number (32)				
Header Length(4) Reserved(6) Code Bits(6)		Window (16)		
Checksum (16)		Urgent (16)		
Options (0 or 32 if any)				
APPLICATION LAYER DATA (Size varies)				

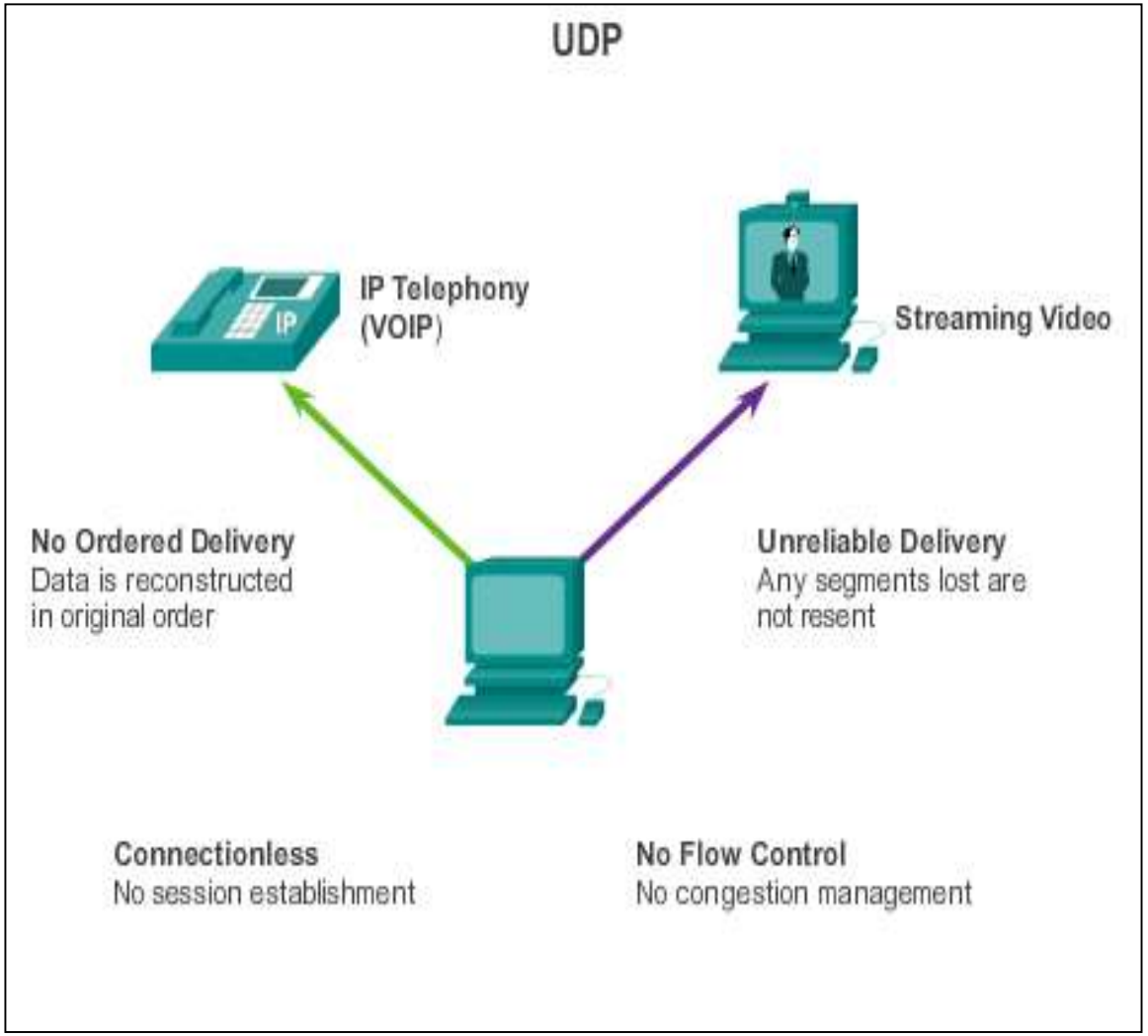
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Introducing UDP

- RFC 768
- Connectionless
- Unreliable delivery
- No ordered data reconstruction
- No flow control
- Stateless protocol

Applications that use UDP:

- Domain Name System (DNS)
- Video Streaming
- VoIP

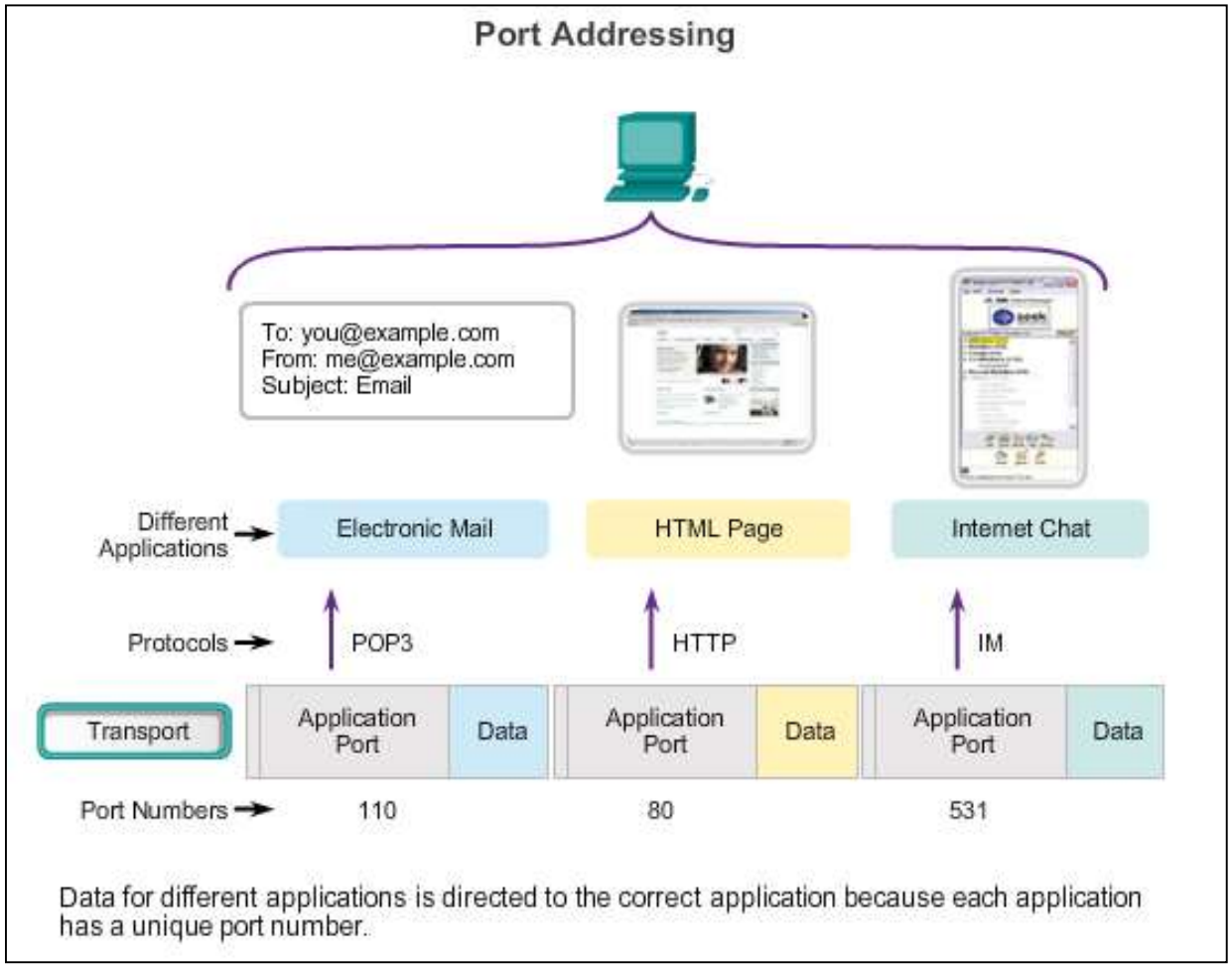


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Separating Multiple Communications

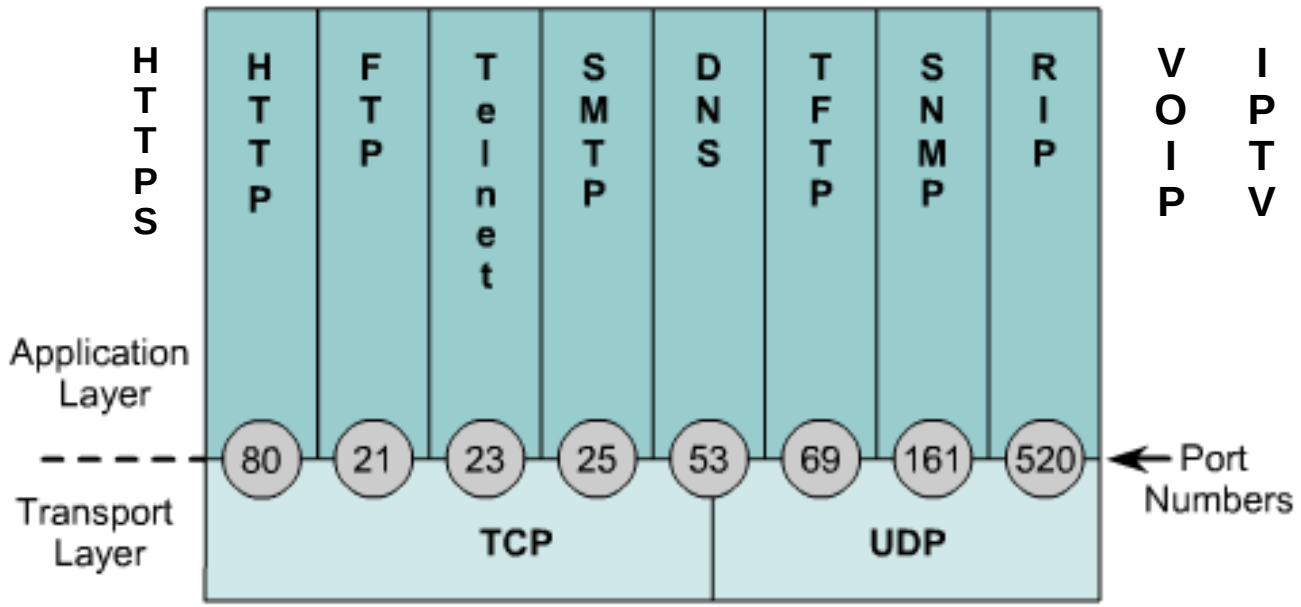
TCP and UDP use port numbers to differentiate between applications.

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Port Numbers



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UDP Low Overhead vs. Reliability

UDP

- Simple protocol that provides the basic transport layer function
- Used by applications that can tolerate small loss of data
- Used by applications that cannot tolerate delay

Used by

- DNS
- Simple Network Management Protocol (SNMP)
- Dynamic Host Configuration Protocol (DHCP)
- Trivial File Transfer Protocol (TFTP)
- IP telephony or VoIP
- Online games

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TCP and UDP Port Addressing (Cont.)

Netstat is used to examine TCP connections that are open and running on a networked host.

```
C:\>netstat

Active Connections

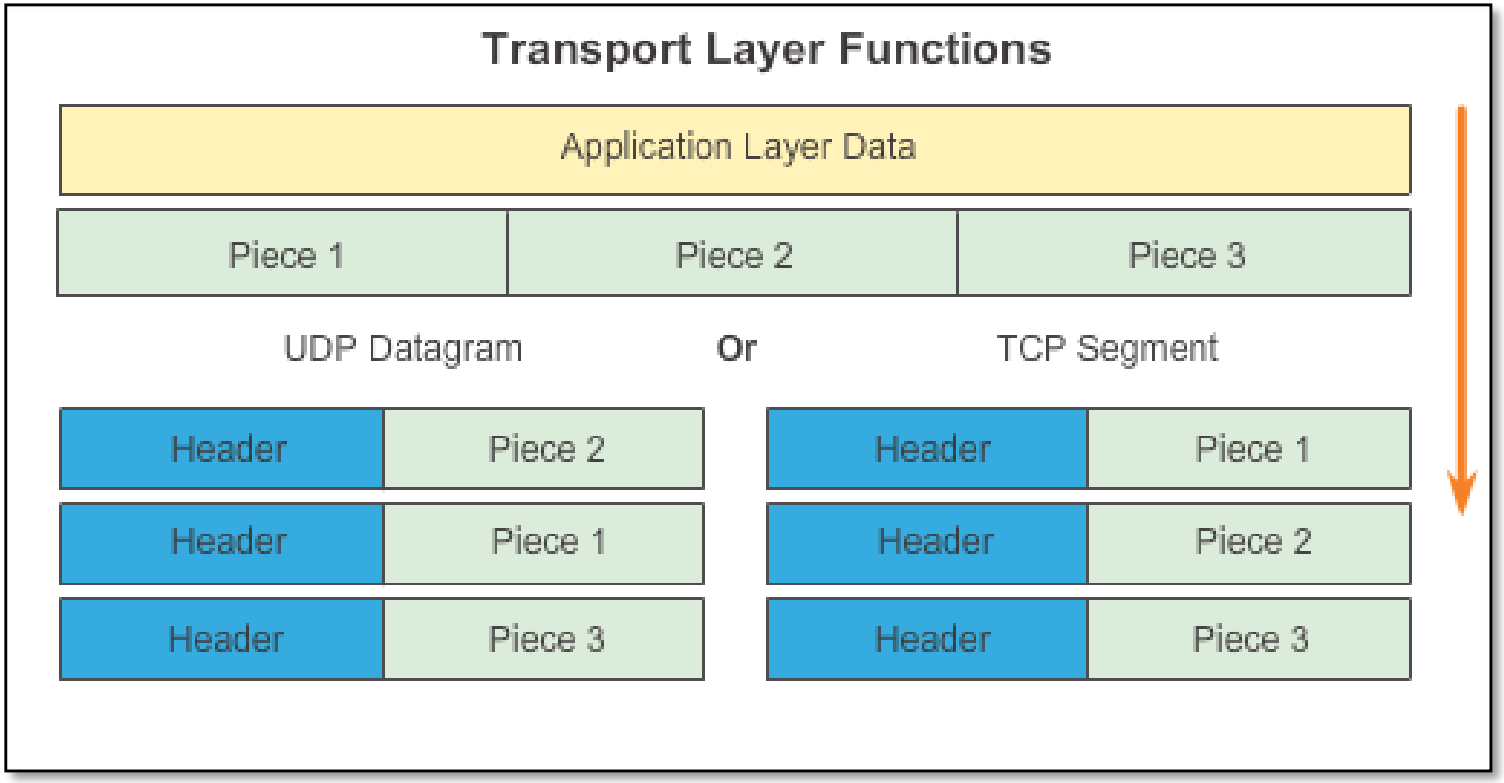
Proto   Local Address    Foreign Address          State
TCP     kenpc:3126       192.168.0.2:netbios-ssn  ESTABLISHED
TCP     kenpc:3158       207.138.126.152:http     ESTABLISHED
TCP     kenpc:3159       207.138.126.169:http     ESTABLISHED
TCP     kenpc:3160       207.138.126.169:http     ESTABLISHED
TCP     kenpc:3161       sc.msn.com:http          ESTABLISHED
TCP     kenpc:3166       www.cisco.com:http       ESTABLISHED

C:\>
```


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TCP and UDP Segmentation

The transport layer divides the data into pieces and adds a header for delivery over the network



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7.2 TCP and UDP



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TCP Connection, Establishment and Termination

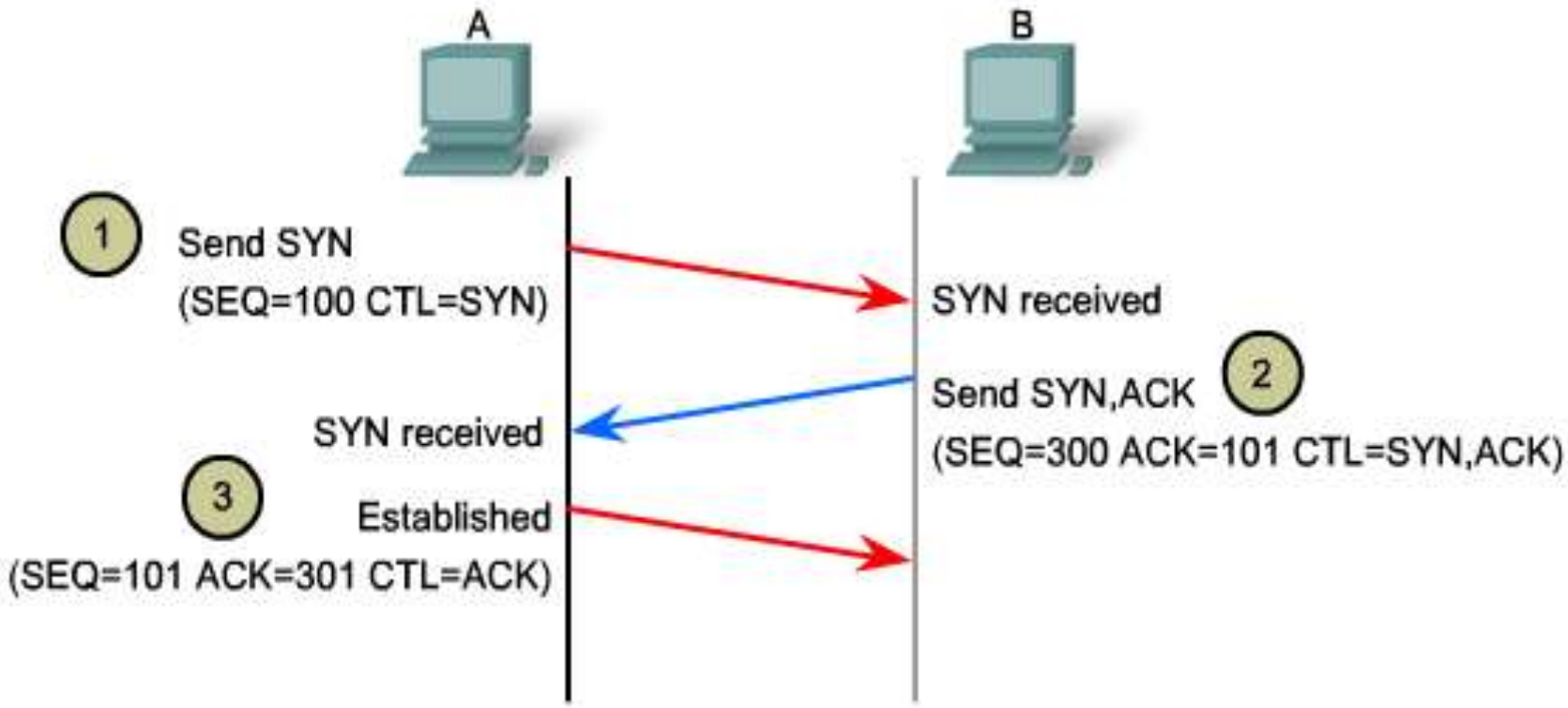
Three-Way Handshake

- Establishes that the destination device is present on the network
- Verifies that the destination device has an active service and is accepting requests on the destination port number that the initiating client intends to use for the session
- Informs the destination device that the source client intends to establish a communication session on that port number

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Three-way Handshake

Starting a TCP Session

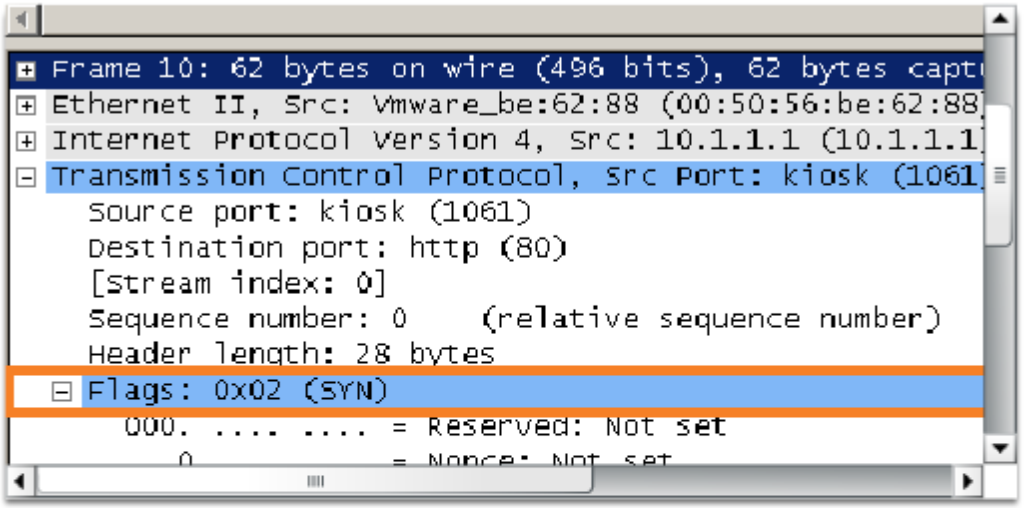


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TCP Three-Way Handshake – Step 1

Step 1: The initiating client requests a client-to-server communication session with the server

TCP 3-Way Handshake (SYN)



A protocol analyzer shows initial client request for session in frame 10

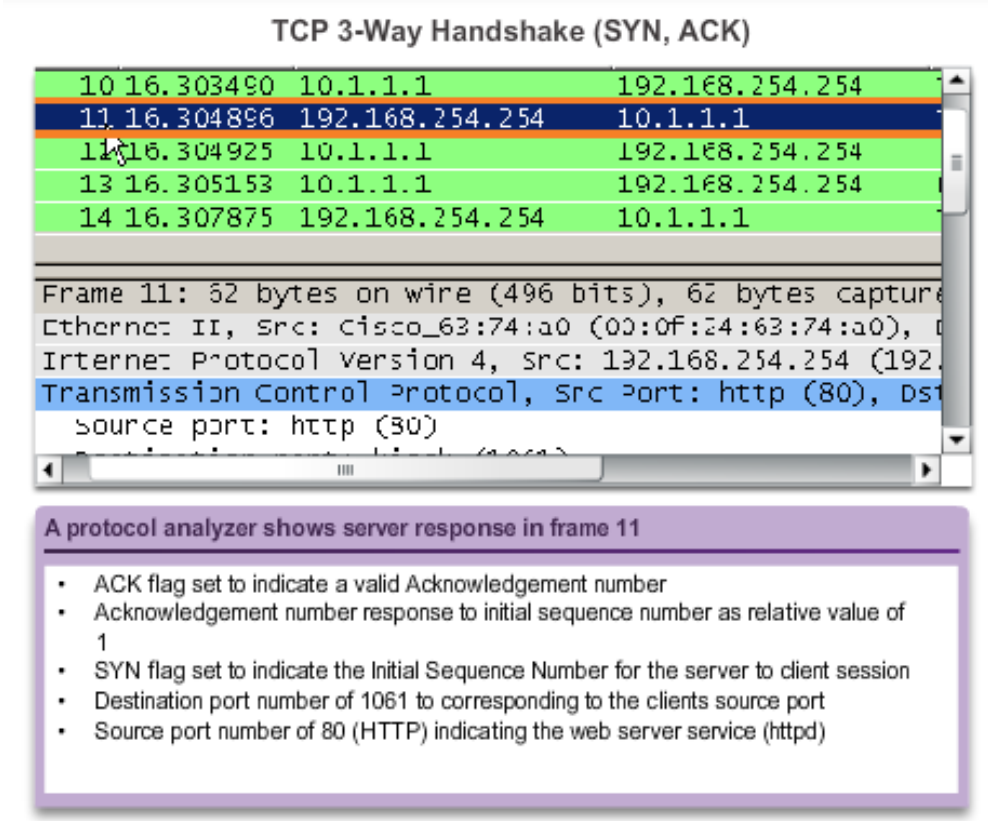
TCP segment in this frame shows:

- SYN flag set to validate an Initial Sequence Number
- Randomized sequence number valid (relative value is 0)
- Random source port 1061
- Well-known destination port is 80 (HTTP port) indicates web server (httpd)

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TCP Three-Way Handshake – Step 2

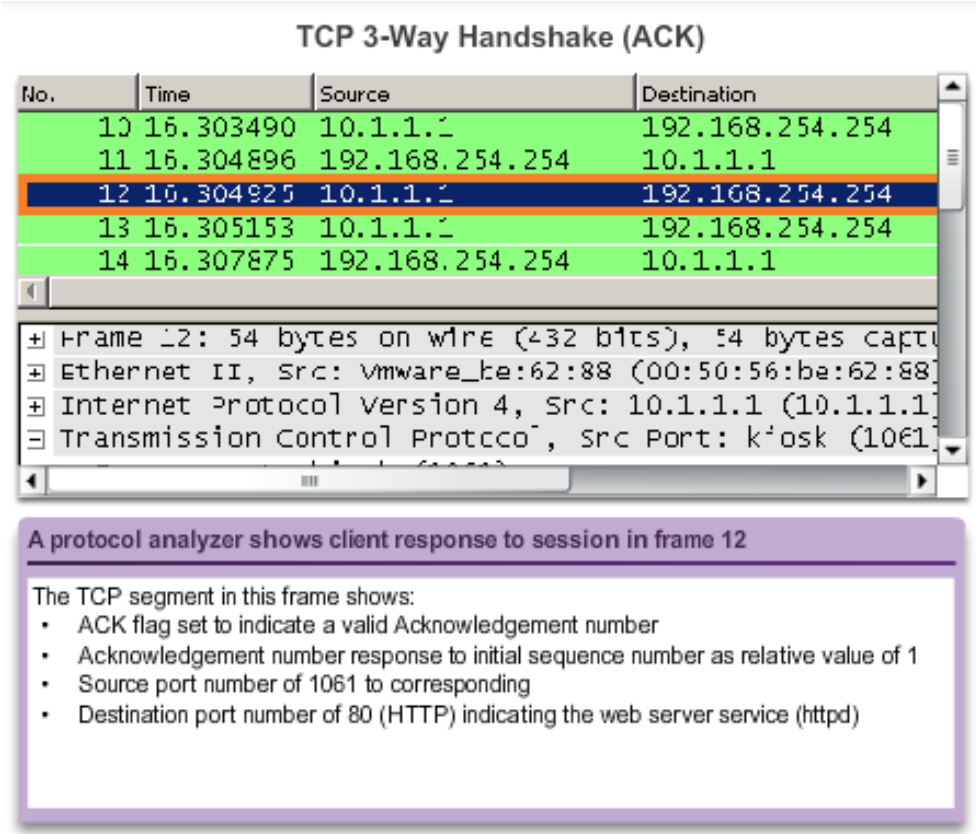
Step 2: The server acknowledges the client-to-server communication session and requests a server-to-client communication session.



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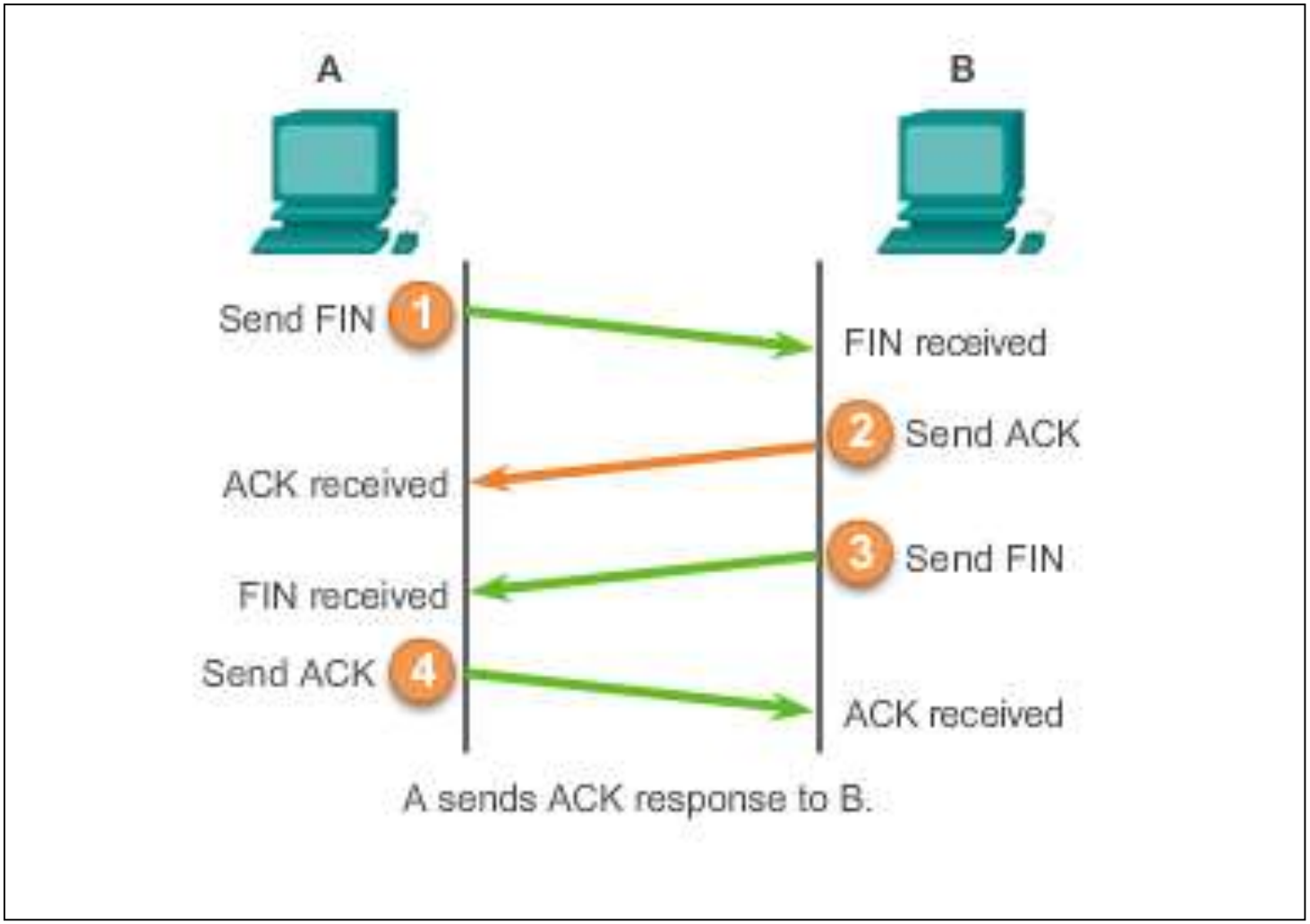
TCP Three-Way Handshake – Step 3

Step 3: The initiating client acknowledges the server-to-client communication session.



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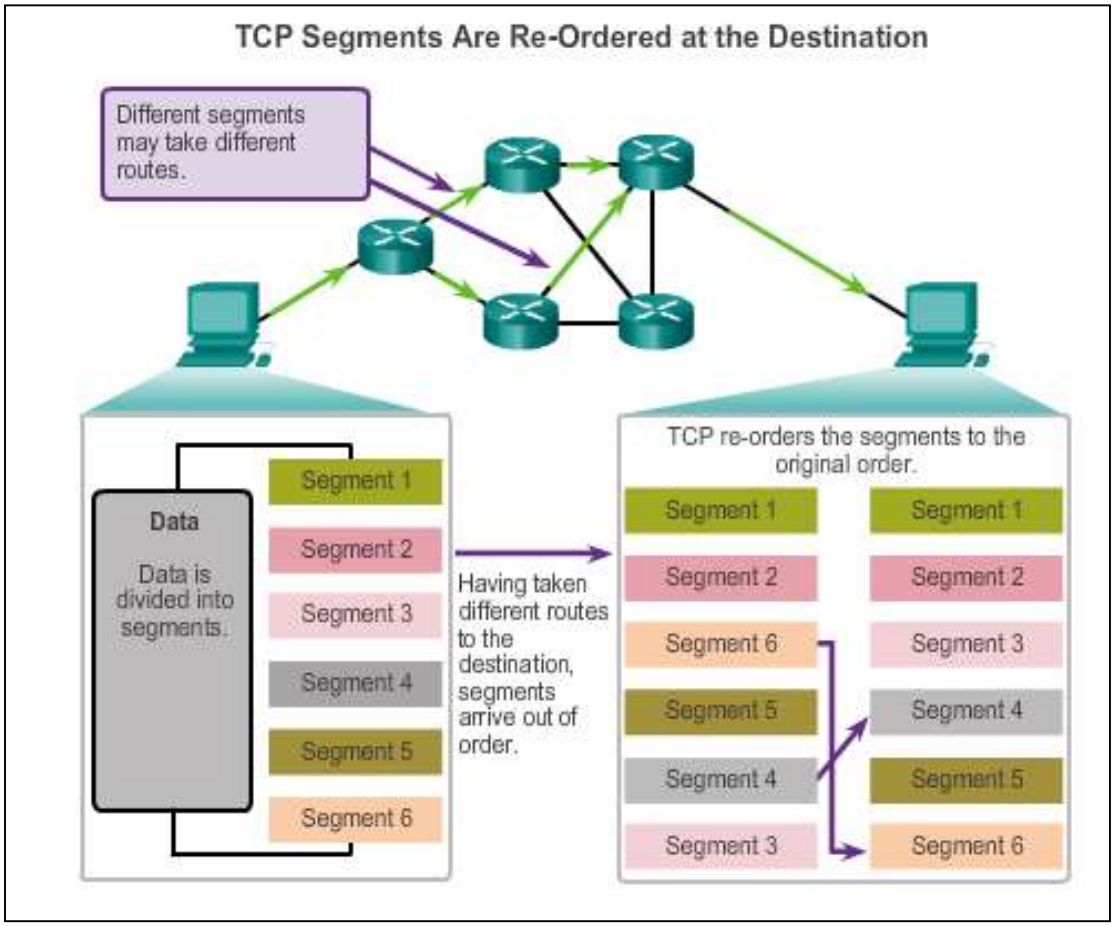
TCP Session Termination



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TCP Reliability – Ordered Delivery

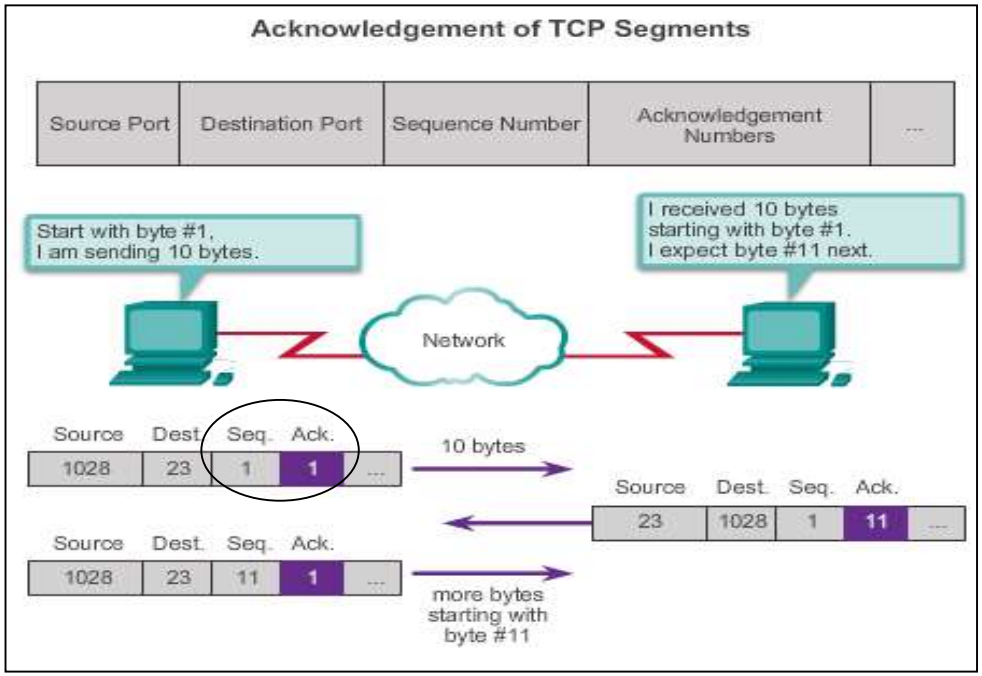
Sequence numbers are used to reassemble segments into their original order.



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Acknowledgement and Window Size

The sequence number and acknowledgement number are used together to confirm receipt.

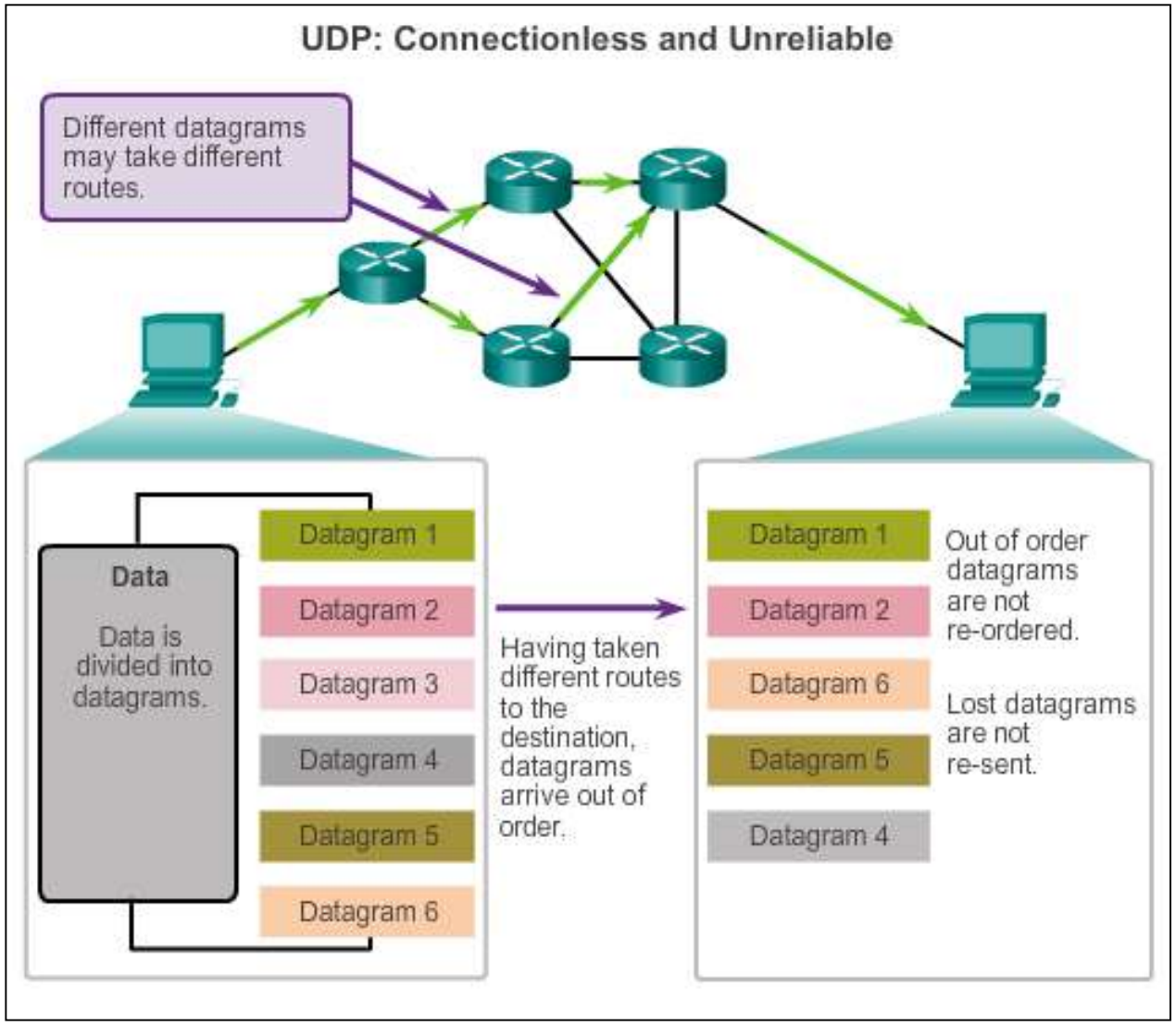


The window size is the amount of data that a source can transmit before an acknowledgement must be received.

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UDP Communication

Datagram Reassembly



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Chapter 7: Summary

In this chapter, you learned:

- The role of the transport layer is to provide three main services: multiplexing, segmentation and reassembly, and error checking. It does this by:
  - Dividing data received from an application into segments.
  - Adding a header to identify and manage each segment.
  - Using the header information to reassemble the segments back into application data.
  - Passing the assembled data to the correct application.
- How TCP and UDP operate and which popular applications use each protocol.
- Transport Layer functions are necessary to address issues in QoS and security in networks.
- Ports provide a “tunnel” for data to get from the transport layer to the appropriate application at the destination.

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